

Luis A. Ramirez

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Education

Yale University <i>Ph.D. in Mechanical Engineering, New Haven, CT</i>	Expected 2027
University of California, Irvine <i>B.S. in Mechanical Engineering, Irvine, CA</i>	2021

Selected Honors, Fellowships, and Awards

Robert E. Apfel Graduate Fellowship	2021
National Science Foundation Graduate Research Fellowship	2021
Engineering Science Symposium Honorable Mention, SHPE National Conference	2020
Edison International Scholarship	2020
UROP Fellowship, University of California, Irvine	2020
SURP Fellowship, University of California, Irvine	2020
UC LEADS Campus Leadership Award, University of California, Irvine	2020
UCI Pathways to Engineering Scholarship, University of California, Irvine	2019
UC LEADS Scholar, University of California, Irvine	2018
First Place Research Poster Presentation, SHPE National Conference	2018
First Place Mechanical Engineering Research Poster, SACNAS National Conference	2017
UC San Diego STARS Scholar	2017

Publications

Peer-Reviewed Journal and Conference Publications

- Ramirez, L. A.**, Baines, R., Yang, B., and Kramer-Bottiglio, R. "Decreasing the Cost of Morphing in Adaptive Morphogenetic Robots." *Advanced Intelligent Systems*, 2401055 (2025).
- Sun, J., Lin, B., **Ramirez, L. A.**, Figueroa, E., Baines, R., Yang, B., Marroquin, E., *et al.* "Performance Enhancement of a Morphing Limb for an Amphibious Robotic Turtle." *Proceedings of the IEEE International Conference on Soft Robotics (RoboSoft)*, pp. 374–379 (2024).
- Baines, R., Yang, B., **Ramirez, L. A.**, and Kramer-Bottiglio, R. "Kirigami Layer Jamming." *Extreme Mechanics Letters*, 102084 (2023).
- Baines, R., Patiballa, S. K., Booth, J., **Ramirez, L. A.**, Sipple, T., Garcia, A., Fish, F., *et al.* "Multi-environment robotic transitions through adaptive morphogenesis." *Nature*, **610**(7931), 283–289 (2022).
- Selvaggio, M., **Ramirez, L. A.**, Naclerio, N. D., Siciliano, B., and Hawkes, E. W. "An Obstacle-Interaction Planning Method for Navigation of Actuated Vine Robots." *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3227–3233 (2020).

Manuscripts in Preparation or Under Review

- Ramirez, L. A.**, Wang, J., Kramer-Bottiglio, R. "When is Shape Change Worth It?" (*In Preparation.*)
- Ramirez, L. A.**, Wang, J., Kramer-Bottiglio, R. "Towards a Deep Water Amphibious Turtle" (*In Preparation.*)

Research Experience

The Faboratory, Yale University <i>Graduate Researcher</i> <i>Advisor: Rebecca Kramer-Bottiglio</i>	2021–Present
Hui Lab, University of California, Irvine <i>Undergraduate Researcher</i> <i>Advisor: Elliot E. Hui</i>	2019–2021
Hawkes Lab, University of California, Santa Barbara	2019–2019

Undergraduate Researcher

Advisor: Elliot W. Hawkes

Biorobotics Laboratory, University of California, Irvine

2018–2019

Undergraduate Researcher

Advisor: David J. Reinkensmeyer

Bioinspired Robotics and Design Laboratory, UC San Diego

2017

Undergraduate Researcher

Advisor: Michael T. Tolley

Selected Presentations

1. **Ramirez, L. A.**, Baines, R., Yang, B., and Kramer-Bottiglio, R. “Decreasing the Cost of Morphing in Adaptive Morphogenetic Robots.” *IEEE International Conference on Soft Robotics (RoboSoft)*, Lausanne, Switzerland, April 2025.
2. **Ramirez, L. A.**, Yang, B., and Kramer-Bottiglio, R. “Robust Multigait Shape Changing Robotic Turtle Platform.” *Machine Learning Automation Robotics Space Conference (MARS)*, Ojai, California, March 2024.
3. **Ramirez, L. A.**, Malhotra, S., Ginoza, G., Zaballa, V., Werner, E., and Hui, E. E. “3-D Printed AMBU Bag for Emergency Bridge Ventilator.” *SHPE National Conference Engineering Science Symposium*, 2020.
4. **Ramirez, L. A.**, Haggerty, D., and Hawkes, E. W. “Vine-Inspired Robot for Endotracheal Intubation.” *UC LEADS Summer Research Presentations*, 2019.
5. **Ramirez, L. A.**, Lijo, A., Sanders, Q., and Reinkensmeyer, D. J. “Design of a Novel Assistive Hand Exoskeleton Using a Residual Grip Force Strategy to Facilitate Pinch Grip.” *SHPE National Conference Engineering Science Symposium*, 2018.
6. **Ramirez, L. A.**, Christianson, C., and Tolley, M. T. “Vertebrate-Inspired Soft Robotic Fish with Dielectric Elastomer Actuators.” *SACNAS National Conference*, 2017.

Mentoring and Teaching

The Laboratory Research Mentoring, Yale University

2023–2026

Graduate Mentor

Ao Jiang , M.S. Student	2025–2026
Kyle Blanset , M.S. Student	2025–2026
Brandon Lin , B.S. Student	2023–2026
Gustavo Zotin Gomes de Oliveria , B.S. Student	2024–2026
Evan Zhang , High school Student	2025
Diana Omar , Undergraduate Researcher	2025
Dylan Sevenikar , B.S. Student	2023

SHPE Graduate Student Advisor, Yale University

2021–2026

Graduate Mentor

- Mentored the SHPE chapter in building its membership, preparing for conferences, and developing professional programming.

UCI Pathways to Engineering Scholarship Program

2019–2021

Mentor

- Mentored undergraduate engineering students in research preparation, résumé development, transfer pathways, internships, scholarships, and graduate-school applications.

Cypress College STEM² Program

2017–2018

Family Group Leader

- Mentored a cohort of 10 community-college STEM students pursuing transfer, research, internship, and scholarship opportunities.

Leadership, Outreach, and Service

Robosoft Workshop Organizer, Shape Morphing Robot Workshop, Robosoft Conference 2024

- Organized a conference workshop, including proposal development, speaker recruitment, and event coordination.

Undergraduate Research Symposium Organizer, Yale University 2023

- Organized a research showcase featuring poster presentations by 10 undergraduate researchers for the Yale robotics community.

Graduate Recruitment Committee Member, Yale University 2023

- Coordinated graduate student recruitment efforts at national diversity conferences by organizing booth logistics and engaging prospective Ph.D. applicants.

Cofounder and Organizer, Mariposa Alumni Scholarship Association 2020–2021

- Helped establish an annual scholarship program and raised approximately \$1,000 for student awards each year.

Marketing Chair, SHPE, University of California, Irvine 2019–2020

- Created a corporate sponsorship package and expanded outreach to students, faculty, and industry partners.

Founding President, SHPE, Cypress College 2017–2018

- Led professional-development workshops serving approximately 70 students per semester and organized engineering outreach activities for underserved communities.

Engineering Mentor and Content Creator, NSF-Funded YouTube Outreach Project 2020

- Created accessible videos about engineering research, higher education, and persistence in STEM for high-school and community-college students.